will be specified when an existing product range is being extended. Suggestions for working principles, however, should always be indicated, in particular when suitable solution principles have emerged during the idea-finding step. These should not prejudice product development (see also the solution-neutral formulation of requirements).

- Indicate a cost target or a budget linked to the company's goals which clarifies future intentions such as production volume, extensions to the product range, new suppliers, etc.

This concludes the product planning phase. By using the listed decision criteria, only those proposals that are likely to fit the company's goals and strengths, and that match the macro- and microeconomic situations, should enter the development stage. The development of the requirements list using the same method that will be applied in product development ensures an easy and seamless transition from product planning to product development.

For successful product planning and development, it is important that both groups work together using the same methods and similar evaluation and decision criteria. At the latest, product development should be actively involved when product ideas are selected and the product is defined. Together they should also develop the requirements list in a format suitable for product development (see Section 5.2).

7. Product Planning in Practice

Because of strong competition, new products have to meet market needs closely, be produced at a competitive cost and be economical to use. In addition, requirements relating to disposal and recycling, and to low environmental impact during production and use, are becoming increasingly important. Products with such complex requirements need to be planned systematically to meet these demands. Just relying on spontaneous ideas or incremental developments to existing products will not, in general, fulfil these demands. Systematic product planning often uses the same methods as concept development, and staff can usefully be exchanged between the two departments.

The following guidelines are important:

- The size of the company determines whether or not it is possible to set up interdisciplinary project groups or departments. In smaller companies it might be necessary to involve external consultants to supply expertise that is missing in the company.

- To use company expertise, however, can involve less risk and often increases client confidence.

- If product planning focuses on existing product lines, in other words further development or systematic variation, the development department responsible for the product line can monitor the new product, or this can be done by a special planning group that includes members from that department.

- When product planning takes place outside an existing product line, in other words the focus is on completely new products or diversification of the product programme, it is better to set up a new planning group. This group works on "innovative planning" and can either be set up as a permanent department or as a temporary working group.
- More elaborate analysis and conscious thought is required when planning for new markets than when dealing with known sales channels and existing client circles.
- When the starting situation is complex, it can be useful to undertake product planning and development using a stepwise and iterative approach. Acquisition of information and the decision making steps should be scheduled such that the anticipated effort and success can be reviewed and planned.
- Even when product ideas have been generated intuitively, a situation analysis and a feasibility study using the search strategies should still be performed.
- To identify customer problems, it is useful to have intensive collaboration with a few leading clients, referred to as "lead users" [3.22]. QFD methods can be used here too [3.11, 3.38].
- When new products are introduced, technical failures and weaknesses can have a far-reaching impact on the reputation of such products. Part of a careful product planning process, therefore, should include sufficient time for testing and the calculation of risks (see Section 7.5.12).
- Entry into the market later than announced can also have a negative effect on reputation because it suggests technical problems.
- During the planning and introduction of new products, it is useful to have a powerful product champion, e.g. a board member who identifies personally with the new product. This helps overcome a potential lack of interest and conventional resistance [3.22].
- Scenario planning (see [3.20, 3.22]) is particularly suitable for long-term forecasts. The effort required for scenario preparation, scenario field analysis, scenario forecasts and scenario building, however, is only worthwhile for business areas that are important to the company and its survival.

Finally, it should be stated that the procedure shown in Figure 3.2 does not represent a straight path with sequential steps, but a guideline for obtaining an essentially purposeful approach. The practical application of this approach will require an iterative procedure in which forward and backward steps at higher levels of information are necessary. This is quite normal in successful product finding.

3.2 Solution Finding Methods

The main advantage of the systematic approach is that designers do not have to rely on coming up with a good idea at the right moment. Solutions can be systematically elaborated using the relevant methods. These methods are the subject of this chapter.

An optimal solution:

- fulfils all demands in the requirements list as well as most of the wishes
- can be realised by the company within the constraints of budget (target costing), time-to-market, production facilities, etc.

Several steps are required to realise such a solution.

First, a range of possible solutions for the given task has to be generated. The basis for this is the function structure (see Section 2.1.3) that is used to divide the overall task into manageable subtasks. The function structure also provides the functional interrelationship between the subtasks, by describing the relationship between the inputs and outputs of each subfunction with respect to the flows of material, energy and signals.

In a second step, one or more possible physical effects are assigned to each of these solution-neutral subfunctions in order to realise them. This is done in accordance with the task-specific requirements. To realise a certain force, for example, a physical effect with the appropriate capability needs to be selected.

The approach described thus far typifies the traditional approach of an engineer. A solution space is created because variants are generated while developing the function structure and when selecting physical effects.

The use of a combination of solution-finding methods can be used to extend the solution space.

Often a subfunction can only be realised through a combination of several physical effects. This is another reason to use several solution finding methods. Those that are proposed or described in the following sections originate from, among others, the area of creativity techniques with its generally recurring methods that are described in Section 2.2.5. Others are based on analogical or logical reasoning.

The methods described here are mainly intended for the design and development of new products. However, they can be very helpful when existing patents of a competitor have to be circumvented or when existing products or components have to be optimised. The methods have to be selected for, adapted to and used in accordance with the context of the problem.

3.2.1 Conventional Methods

1. Information Gathering

For designers, access to state-of-the-art information is essential. As a first step, designers use a variety of collection techniques [3.45]. Information and data repositories, along with systems used to search and process the data, assist the active search for and the passive discovery of solutions. The internet enables a more effective and efficient application of the following conventional techniques:

- searching the literature
- analysing trade publications

- surveying the presentations from exhibitions and fairs
- assessing catalogues of competitors
- exploring patents, etc.

2. Analysis of Natural Systems

The study of natural forms, structures, organisms and processes can lead to very useful and novel technical solutions. The connections between biology and technology are investigated by bionics and biomechanics. Nature can stimulate the creative imagination of designers in a host of different ways [3.6, 3.29, 3.31, 3.35].

Technical applications of the design principles of natural forms include lightweight structures employing honeycombs, tubes and rods, the profiles of aircraft and ships, and the take-off and flying characteristics of aircraft. Lightweight structures in the form of thin stems are very important (see Figure 3.8). Another technical application is sandwich construction, and Figure 3.9 shows a few derivations of this natural principle that have proved useful in aircraft construction.

The hooks of a burr provided a solution that was incorporated into the Velcro fastener (see Figure 3.10). Further examples are given in Figure 3.11.

Fibre composites can be used to optimise the stiffness and deformation of structures that can equal or exceed those in found in nature. Carbon, glass and plastic fibres are aligned according to the principal stress directions and embedded in a predominantly polymer matrix of polyester, epoxy and other resins. This construction method requires an in-depth stress analysis along with a laying-up technique for the fibres adapted to that analysis, as well as extensive knowledge of plastics to select the fibre matrix composite. The basic relationships and ideas for the correct design of fibre composites and numerous literature references are provided by Flemming et al. [3.16].

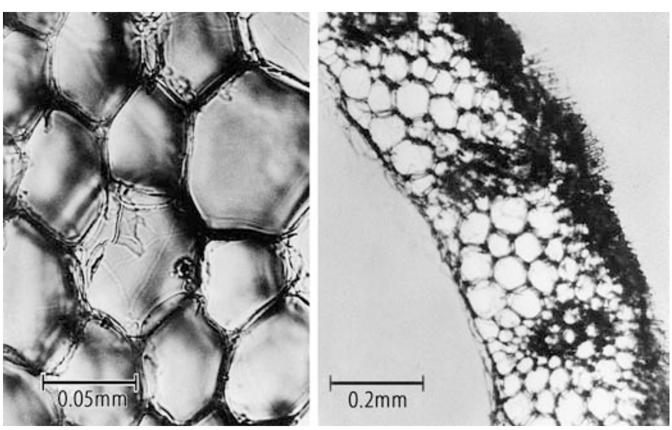


Figure 3.8. Wall of a wheat stem [3.29]

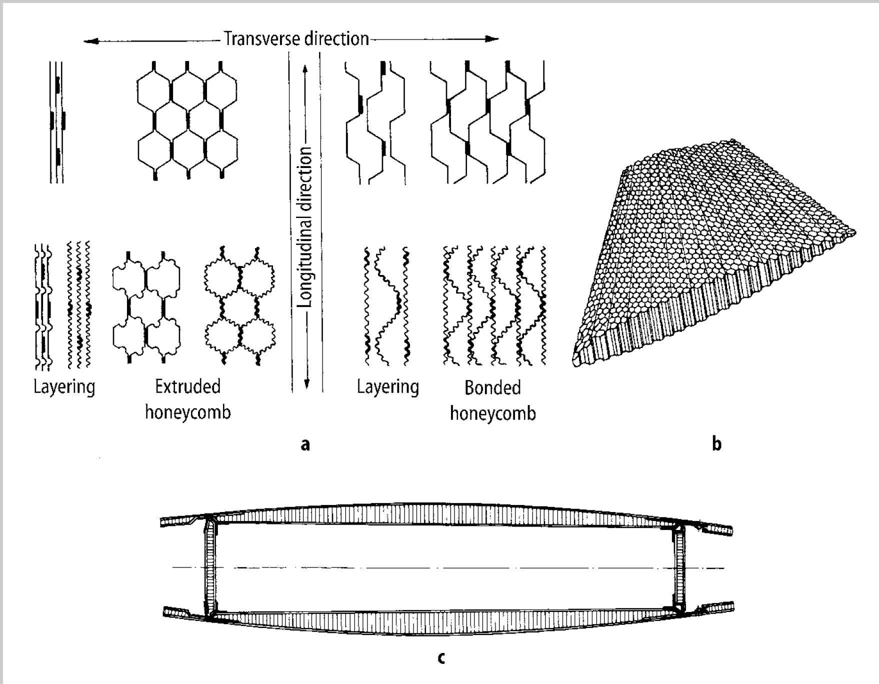


Figure 3.9. Sandwich construction for lightweight structures [3.30]. **a** A few honeycomb structures. **b** Completed honeycomb structure. **c** Sandwich box girder

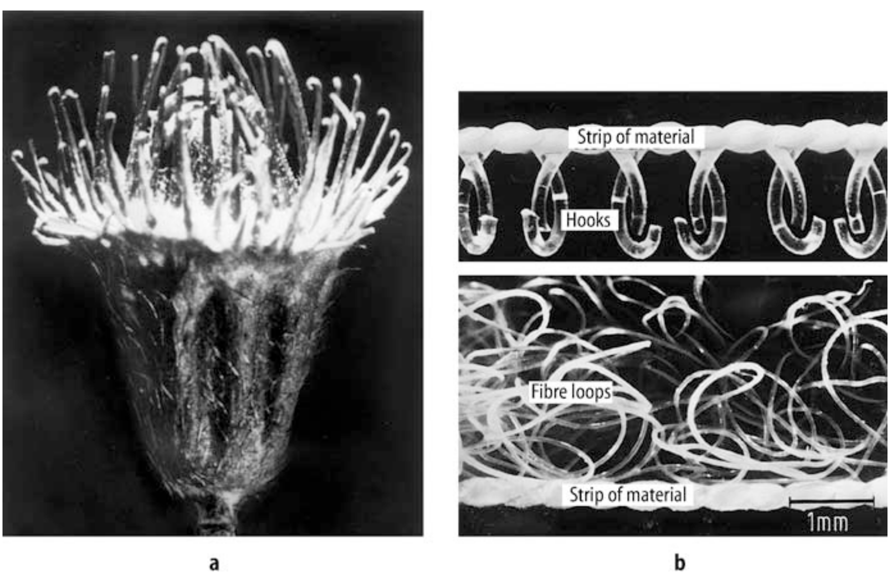


Figure 3.10. a Hooks of a burr. **b** Velcro fastener. After [3.29]

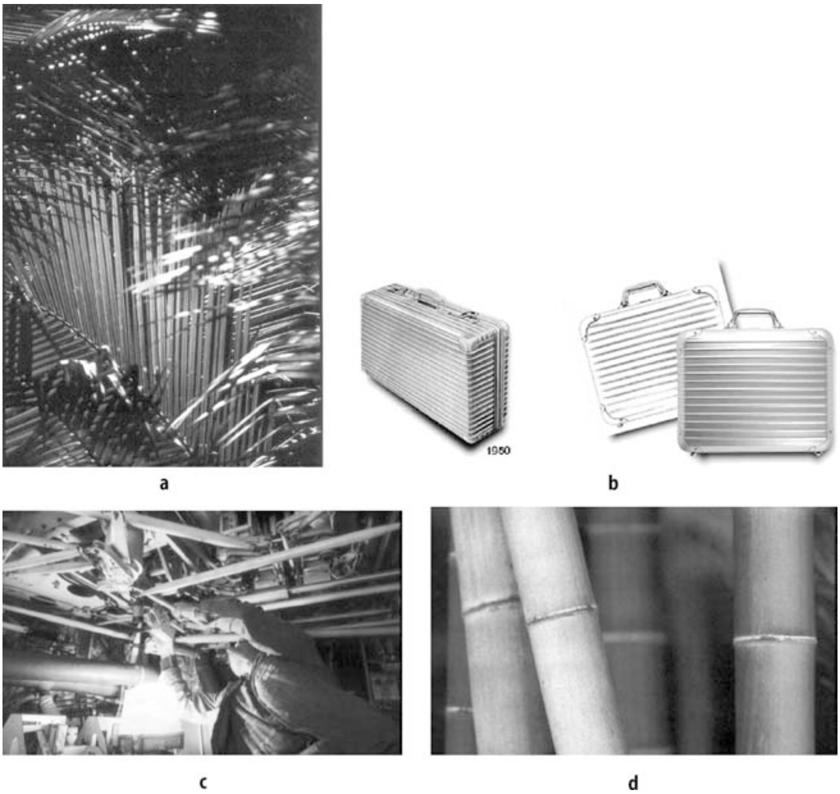


Figure 3.11. **a** Palm leaves (Lufthansa publication 2/96). **b** Aluminium suitcase (Rimowa Kofferfabrik 10/01). **c** Tubular structure in an aircraft. **d** Bamboo stems (Lufthansa publication 5/96)

3. Analysis of Existing Technical Systems

The analysis of existing technical systems is one of the most important means of generating new or improved solution variants in a step-by-step manner.

This analysis involves the mental or even physical dissection of finished products. It may be considered a form of structure analysis (see Subsection 1 in Section 2.2.5) aimed at the discovery of related logical, physical and embodiment design features. Figure 6.10 shows an example of this type of analysis. Here, subfunctions were derived from the existing configuration. From them, further analysis led to the identification of the physical effects involved which, in turn, might have suggested new solution principles for corresponding subfunctions. It is also possible to adopt solution principles discovered during the analysis.

Existing systems used for analysis might include:

- products or production methods from competing companies
- older products and production methods from one's own company
- similar products or assemblies in which some subfunctions or parts of the function structure correspond to those for which a solution is being sought.

Because the only systems to be analysed are those that have some bearing on the new problem as a whole or on parts of it, we could call this way of collecting information the systematic exploitation of proven ideas, or of experience. It proves particularly helpful for finding a first solution concept as a starting point for further variations. It must, however, be said that this approach carries the danger of causing designers to stick with known solutions instead of pursuing new paths.

4. Analogies

In the search for solutions and in the analysis of system properties, it is often useful to substitute an analogous problem (or system) for the one under consideration, and to treat it as a model. In technical systems, analogies may be obtained, for instance, by changing the type of energy used [3.3, 3.64]. Analogies chosen from the nontechnical sphere may prove very useful as well.

Besides helping in the search for solutions, analogies are also helpful in the study of the behaviour of a system during the early stages of its development by means of simulation and model techniques, and in the subsequent identification of essential new subsolutions and the introduction of early optimisations.

If the model is to be applied to systems of markedly different dimensions and conditions, a supportive similarity (dimensional) analysis should be undertaken (see Section 9.1.1).

5. Measurements and Model Tests

Measurements on existing systems, model tests supported by similarity analyses and other experimental studies are among the most important sources of information. Rodenacker [3.59] in particular stresses the importance of experimental studies, arguing that design can be interpreted as the reversal of physical experiment.

In the precision engineering and mass production industries, including those where micromechanisms and electronic products are developed, experimental investigations are an important and established means of arriving at solutions. This approach has organisational repercussions since, in the creation of such products, experimental development is often incorporated within the design activity (see Figure 1.3).

In a similar way, the testing and subsequent modification of software solutions belong to this empirically based group of methods.

3.2.2 Intuitive Methods

Designers often seek and discover solutions for difficult problems by intuition—that is, solutions come to them in a flash after a period of search and reflection. These solutions suddenly appear as conscious thoughts and often their origins cannot be traced. As Galtung of the International Peace Research Institute in Oslo has put it: “The good idea is not discovered or undiscovered; it comes, it happens”.

It is then developed, modified and amended, until such time as it leads to the solution of the problem.

Good ideas are always scrutinised by the subconscious or preconscious in the light of expert knowledge, experience and the task in hand, and often the simple impetus resulting from the association of ideas suffices to force them into consciousness. That impetus can also come from apparently unconnected external events or discussions. Frequently, a sudden idea will hit the bull's eye, so that all that needs to be done is to make changes or adaptations that lead straight to a final solution. If that is indeed the case and a successful product is created, then this represents the optimum procedure. Very many good solutions are born in that way and developed successfully. A good design method, far from trying to eliminate this process, should serve to back it up.

An industrial concern should nevertheless beware of exclusive reliance on the intuition of its designers, nor should designers themselves leave everything to chance or rare inspiration. Purely intuitive methods have the following disadvantages:

- The right idea does not always come at the right time, since it cannot be forced.
- Current conventions and personal prejudices may inhibit original developments.
- Because of inadequate information, new technologies or procedures may fail to reach the consciousness of the designer.

These dangers increase with specialisation, the division of tasks and with time pressure.

There are several methods of encouraging intuition and opening new paths by the association of ideas. The simplest and most common of these involves critical discussions with colleagues. Provided that such discussions are not allowed to stray too far and are based on the general methods of persistent questions, negation, forward steps, etc. (see Section 2.2.5), they can be very helpful and effective.

Methods with an intuitive bias such as Brainstorming, Synectics, Gallery Method, Method 635 and many others involve group dynamics that are used to generate the widest possible range of ideas. One of the effects of group dynamics is the uninhibited exchange of associated ideas between the members.

Most of these techniques were originally devised for the solution of nontechnical problems. They are, however, applicable to any field that demands new, unconventional ideas.

1. Brainstorming

Brainstorming can be described as a method of generating a flood of new ideas. It was originally suggested by Osborn [3.51] and provides conditions in which a group of open-minded people from as many different spheres of life as possible bring up, without prejudice, any thoughts that occur to them and thus trigger off new ideas in the minds of the other participants [3.74]. Brainstorming relies strongly on stimulation of the memory and on the association of ideas that have never been considered in the current context or have never been allowed to reach consciousness.

For maximum effect, brainstorming sessions should be run along the following lines:

Composition of the Group

- The group should have a leader and consist of a minimum of five and a maximum of 15 people. Fewer than five constitute a spectrum of opinion and experience that is too small, and hence produce too few stimuli. With more than 15, close collaboration may decline because of individual passivity and withdrawal.
- The group must not be confined to experts. It is important that as many fields and activities as possible are represented, the involvement of nontechnical members adding a rich new dimension.
- The group should not be hierarchically structured but, if possible, made up of equals in order to prevent the censoring of such thoughts as might give offence to superiors or subordinates.

Leadership of the Group

- The leader of the group should only take the initiative when dealing with organisational problems (invitation, composition, duration and evaluation). Before the actual brainstorming session, the leader must outline the problem and, during the session, must see to it that the rules are observed and, in particular, that the atmosphere remains free and easy. To that end the leader should start the session by expressing a few absurd ideas, or mentioning an example from another brainstorming session, but should never lead in the expression of ideas. On the other hand, the flow of new ideas should be encouraged whenever the productivity of the group slackens. The leader must ensure that no one criticises the ideas of other participants, and should appoint one or two members to take minutes.

Procedure

- All participants must try to shed their intellectual inhibitions; that is, they should avoid rejecting as absurd, false, embarrassing, stupid, well-known or redundant any ideas expressed spontaneously by themselves or by other members of the group.
- No participant should criticise any ideas that are brought up, and everyone must refrain from using such killer phrases as “we’ve heard it all before”, “it can’t be done”, “it will never work” and “this has nothing to do with the problem”.
- New ideas will be taken up by the other participants, who may change and develop them at will. It is also useful to combine several ideas into new proposals.
- All ideas should be written down, sketched out, or recorded.
- All suggestions should be concrete enough to allow the emergence of specific solution ideas.
- The practicability of the suggestions should be ignored at first.

- A session should not generally last for more than 30 to 45 minutes. Experience has shown that longer sessions produce nothing new and lead to unnecessary repetitions. It is better to make a fresh start with new ideas or with other participants later.

Evaluation

- The results should be reviewed by experts to find potential solution elements. If possible, these should be classified and graded in order of feasibility and then developed further.
- The final result should be reviewed with the entire group to avoid possible misunderstandings or one-sided interpretations on the part of the experts. New and more advanced ideas may well be expressed or developed during such a review session.

Brainstorming is indicated [3.56] whenever:

- No practical solution principle has been discovered.
- The physical process underlying a possible solution has not yet been identified.
- There is a general feeling that deadlock has been reached.
- A radical departure from the conventional approach is required.

Brainstorming is even useful in the solution of subproblems arising in known or existing systems. Moreover, it has a beneficial side-effect: all of the participants are supplied with new data, or at least with fresh ideas on possible procedures, applications, materials, combinations, etc., because the group represents a broad spectrum of opinion and expertise (for instance, designers, production engineers, sales persons, materials experts and buyers). It is astonishing what a profusion and range of ideas such a group can generate. The designers will remember the ideas brought up during brainstorming sessions on many future occasions. Brainstorming triggers off new lines of thought, stimulates interest and represents a break in the normal routine.

It should, however, be stressed that no miracles must be expected from brainstorming sessions. Most of the ideas expressed will not be technically or economically feasible, and those that are will often be familiar to the experts. Brainstorming is meant first of all to trigger off new ideas, but it cannot be expected to produce ready-made solutions because problems are generally too complex and too difficult to be solved by spontaneous ideas alone. However, if a session should produce one or two useful new ideas, or even some hints in what direction to go looking for the solution, it will have achieved a great deal.

An example of a solution obtained by Brainstorming can be found in Section 6.6, which also shows how the resulting ideas were evaluated and how classifying criteria for the subsequent search for solutions were derived from them.

2. Method 635

Brainstorming has been developed into Method 635 by Rohrbach [3.60]. After familiarising themselves with the task, and after careful analysis, each of six participants is asked to write down three rough solutions in the form of keywords. After some time, the solutions are handed to each participant's neighbour who, after reading the previous suggestions, enters three further solutions or developments. This process is continued until each original set of three solutions has been completed or developed through association by the five other participants, hence the name of the method.

Method 635 has the following advantages over Brainstorming:

- A good idea can be developed more systematically.
- It is possible to follow the development of an idea and to determine more or less reliably who originated the successful solution principle, which might prove advisable for legal reasons.
- The problem of group leadership rarely arises.

The method has the following disadvantage:

- Reduced creativity by the individual participants owing to isolation, and lack of stimulation in the absence of overt group activity.

3. Gallery Method

The Gallery Method developed by Hellfritz [3.27] combines individual work with group work, and is particularly suitable for any stage of the design process where solution proposals can be expressed in the form of sketches or drawings. The organisation and team building are similar to Brainstorming. The method consists of the following steps.

Introduction Step: The group leader presents the problem and explains the context.

Idea Generation Step 1: For 15 minutes the individual group members create solutions intuitively and without prejudice using sketches supported, where necessary, by text.

Association Step: The results from idea generation step 1 are hung on a wall as in an art gallery so that all group members can see and discuss them. The purpose of this 15-minute association step is to find new ideas or to identify complementary or improved proposals through negation and reappraisal.

Idea Generation Step 2: The ideas and insights from the association step are further developed individually by each of the group members.

Selection Step: All ideas generated are reviewed, classified and, if necessary, finalised. Promising solutions are then selected (see Section 3.3.1). It is also possible to identify potential solution characteristics that can be developed later using a discursive method (see Section 3.2.3).

The Gallery Method has the following advantages:

- Intuitive group working takes place without unduly lengthy discussions.
- An effective exchange of ideas using sketches is possible.
- Individual contributions can be identified.
- Documentary records are easily assessed and stored.

4. Delphi Method

In this method, experts in a particular field are asked for written opinions [3.7].

The requests take the following form:

First Round: What starting points for solving the given problem do you suggest? Please make spontaneous suggestions.

Second Round: Here is a list of various starting points for solving the given problem. Please go through this list and make what further suggestions occur to you.

Third Round: Here is the final evaluation of the first two rounds. Please go through the list and write down what suggestions you consider most practicable.

This elaborate procedure must be planned very carefully and is usually confined to general problems bearing on fundamental questions or on company policy. In the field of engineering design, the Delphi Method should be reserved for fundamental studies of long-term developments.

5. Synectics

Synectics is a word derived from Greek and it refers to the activity of combining various and apparently independent concepts. Synectics is comparable to Brainstorming, with the difference that its aim is to trigger off fruitful ideas with the help of analogies from nontechnical or semi-technical fields.

The method was first proposed by Gordon [3.25]. It is more systematic than Brainstorming, with its arbitrary flow of ideas. However, both methods call for complete frankness and lack of inhibition or criticism.

A synectics group should consist of no more than seven members, otherwise the ideas expressed will run away with themselves. The leader of the group has the additional task of helping the group to develop the proposed analogies by guiding them through the following steps:

- Presentation of the problem.
- Familiarisation with the problem (analysis).
- Grasping the problem.
- Rejection of familiar assumptions with the help of analogies drawn from other spheres.
- Analysis of one of the analogies.
- Comparison of the analogy with the existing problem.
- Development of a new idea from that comparison.
- Development of a possible solution.

If the result is unsatisfactory, the process may have to be repeated with a different analogy.

An example may help to illustrate this method. In a seminar set up for the purpose of discovering the best method of removing urinary calculi from the human body, several mechanical devices for gripping, holding and extracting these stones were mentioned. The device would have to stretch and open up inside the urethra. The keywords "stretch" and "open up" suggested the idea of an umbrella to one of the participants (see Figure 3.12).

Question: How can the umbrella analogy—(a) in Figure 3.12—be applied?

Possible answer 1: By (b) drilling through the stone, pushing the umbrella through the hole and opening it up. Not very feasible.

Possible answer 2: By (c) pushing a tube through the hole and blowing it up (balloon) behind the stone. Drilling of hole not feasible.

Possible answer 3: By (d) pushing the tube past the stone. When the tube is withdrawn the resistance may seriously damage the urethra.

Possible answer 4: By (e) adding a second balloon as a guide and by (f) embedding the stone in a gel between the two balloons and then pulling it out? This was found to be the best solution.

This example shows the association with a semi-technical analogy (umbrella) from which a solution was developed that took into account the special constraints that existed in this case. The solution shown here is not the final solution resulting from the seminar but represents an example of how the method was used.

Characteristic of this approach is the unrestricted use of analogies which, in the case of technical problems, are selected from nontechnical or semi-technical spheres. Such analogies will generally suggest themselves quite spontaneously at the first attempt but, during subsequent development and analysis, they will generally be derived more systematically.

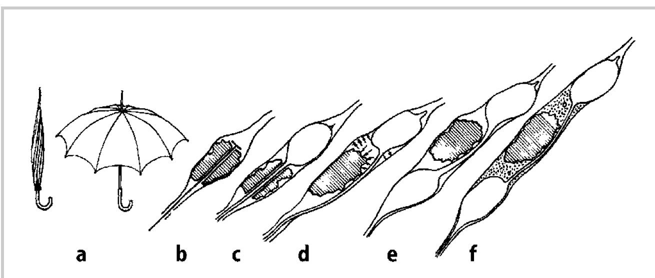


Figure 3.12. Step-by-step development of a solution principle for the removal of urinary calculi based on analogy and stepwise improvement

6. Combination of Methods

Any one of these methods taken by itself may not lead to the required goal. Experience has shown that:

- The group leader of, or another participant in, a brainstorming session may, when the flow of ideas dries up, introduce synectic procedures—deriving analogies, rejection of familiar assumptions, etc.—to release a new flood of ideas.

- A new idea or an analogy may radically change the approach and ideas of the group.
- A summary of what has been agreed so far may lead to new ideas.
- The explicit use of the methods of negation and reappraisal and of forward steps (see Section 2.2.5) can enrich and extend the variety of ideas.

In the seminar we mentioned, the presentation of the idea "destroy stone" produced a host of new suggestions, such as drilling, smashing, hammering, ultrasonic disintegration and so on. When the flow of ideas eventually dried up, the group leader asked, "How does nature destroy?", which immediately evoked a number of new suggestions, including weathering, heating and cooling, decay, putrefaction, bacterial action, ice expansion and chemical decomposition. A combination of the two principles "clasp stone" and "destroy stone" provoked the question, "What else?" This produced the answer "contact stone rather than clasp", which in turn threw up such new ideas as sucking, gluing, and applying various contact forces.

The different methods should be combined so as to best address particular cases. A pragmatic approach ensures the best results.

3.2.3 Discursive Methods

Methods with a discursive bias provide solutions in a deliberate step-by-step approach that can be influenced and communicated. Discursive methods do not exclude intuition, which can make its influence felt during individual steps and in the solution of individual problems, but not in the direct implementation of the overall task.

1. Systematic Study of Physical Processes

If the solution of a problem involves a known physical (chemical, biological) effect represented by an equation, and especially when several physical variables are involved, various solutions can be derived from the analysis of their interrelationships, that is, of the *relationship* between a dependent and an independent variable, all other quantities being kept constant. Thus, if we have an equation in the form $y = f(u, v, w)$, then, according to this method, we investigate solution variants for the relationships $y_1 = f(u, \underline{v}, \underline{w})$, $y_2 = f(\underline{u}, v, \underline{w})$ and $y_3 = f(\underline{u}, \underline{v}, w)$, the underlined quantities being kept constant.

Rodenacker has given several examples of this procedure, one of which concerns the development of a capillary viscometer [3.59]. Four solution variants can be derived from the well-known law of capillary action $\eta \sim \Delta p \cdot r^4/(\dot{V} \cdot l)$. They are shown schematically in Figure 3.13.

1. A solution in which the differential pressure Δp serves as a measure of the viscosity: $\Delta p \sim \eta$ ($\dot{V}$, r and l = constant).
2. A solution based on changes in radius of the capillary tube: $\Delta r \sim \eta$ ($\dot{V}$, Δp and l = constant).

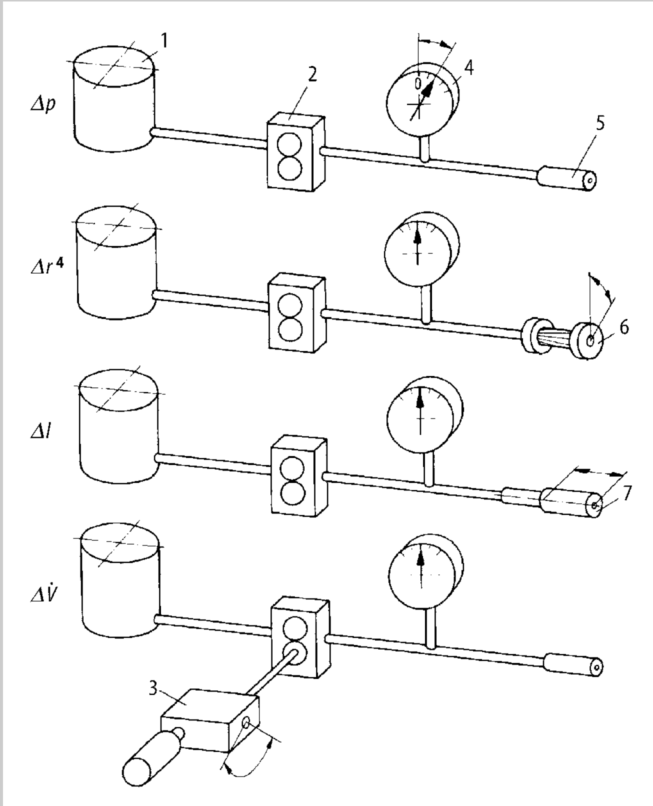


Figure 3.13. Schematic representation of four viscometers, after [3.59]. *1*, container; *2*, gear pump; *3*, variable drive; *4*, pressure gauge; *5*, fixed capillary tube; *6*, capillary tube with variable diameter; *7*, capillary tube with variable length

3. A solution based on changes in the length of the capillary tube: $\Delta l \sim \eta$ (Δp, $\dot{V}$ and r = constant).
4. A solution based on changes in the volume flow rate: $\Delta\dot{V} \sim \eta$ (Δp, r and l = constant).

Another way of obtaining new or improved solutions by the analysis of physical equations is the resolution of known physical effects into their individual components. Rodenacker [3.59], in particular, has used this approach in the design of novel devices and the development of new applications for existing ones.

By way of example, let us look at the development of a frictional thread locking device, based on the analysis of the equation governing the torque needed to release a threaded fastener:

$$T = P[(d/2)\tan(\phi_v - \beta) + (D/2)\mu_f] \tag{3.1}$$

The torque given by Equation (3.1) is made up of the following components:

Frictional torque in the thread:

$$T_t \sim P(d/2)\tan\phi_v = P(d/2)\mu_v \tag{3.2}$$

where

$$\tan\phi_v = \mu_t/\cos(\alpha/2) = \mu_v$$

Frictional torque on the bolt head or nut face:

$$T_f = P(D/2)\tan\phi_f = P(D/2)\mu_f \tag{3.3}$$

Release torque of the thread due to pre-load and thread pitch:

$$T_r \sim P(d/2)\tan(-\beta) = -P \cdot \frac{P}{2\pi} \tag{3.4}$$

(where p = thread pitch, β = helix angle, d = mean thread (t) diameter, P = pre-load, D = mean face (f) diameter, μ_v = virtual (v) coefficient of friction in the thread, μ_t = actual coefficient of friction in the thread, μ_f = coefficient of friction on the head or nut face, α = flank angle, ϕ = angle of friction).

To discover solution principles for the improvement of the locking properties of a threaded fastener, we must analyse the physical relationships further so as to identify the physical effects involved. The individual effects involved in Equations (3.2) and (3.3) are:

- the friction effect (Coulomb friction)

$$F_t = \mu_v P \quad \text{and} \quad F_f = \mu_f P$$

- the lever effect

$$T_t = F_t\, d/2 \quad \text{and} \quad T_f = F_f\, D/2$$

- the wedge effect

$$\mu_v = \mu_t/\cos(\alpha/2)$$

The individual effects in Equation (3.4) are:

- the wedge effect

$$F_r \sim P\tan(-\beta)$$

- the lever effect

$$T_r = F_r\, d/2$$

An examination of the individual physical effects will yield the following solution principles for the improvement of the locking properties of the fastener:

- Use of the wedge effect to reduce the tendency to loosen by decreasing the helix angle β.

- Use of the lever effect to increase the frictional moment on the head or nut face by increasing the mean face diameter D.
- Use of the friction effect to increase the frictional force by increasing the coefficient of friction μ.
- Use of the wedge effect to increase the frictional force on the face by means of conical surfaces ($P\mu_f/\sin\gamma$ with cone angle $= 2\gamma$). This method is used with automobile wheel attachment nuts.
- Increase of the flank angle α to increase the virtual coefficient of friction in the thread.

2. Systematic Search with the Help of Classification Schemes

In Section 2.2.5 we showed that the systematic presentation of information and data is helpful in two respects. On the one hand it stimulates the search for further solutions in various directions; on the other hand it facilitates the identification and combination of essential solution characteristics. Because of these advantages, a number of classification schemes have been drawn up, all with a similar basic structure. Dreibholz [3.10] has published a comprehensive survey of the possible applications of such classification schemes.

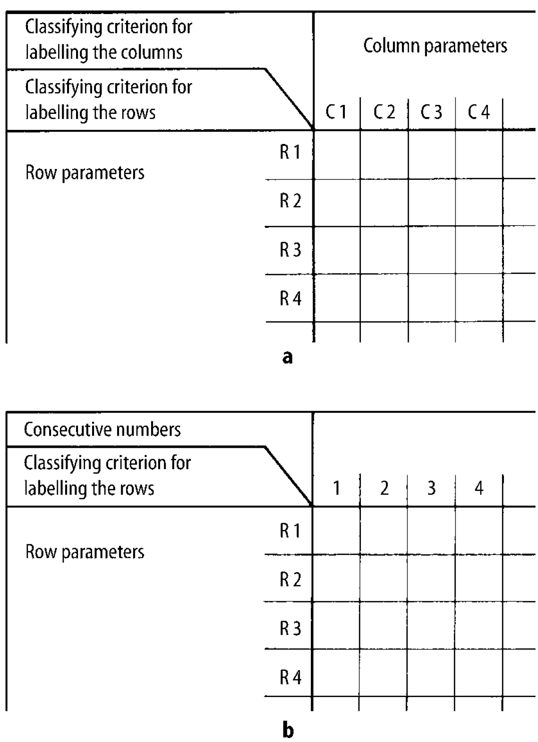


Figure 3.14. General structure of classification schemes. After [3.10]

The usual two-dimensional scheme consists of rows and columns of parameters used as classifying criteria. Figure 3.14 illustrates the general structure of classification schemes: (a) when parameters are provided for both the rows and the columns; and (b) when parameters are provided for the rows only, because the columns cannot be arranged in any apparent order. If necessary, the classifying criteria can be extended by a further breakdown of the parameters or characteristics (see Figure 3.15); this, however, often tends to confuse the general picture. By allocating the column parameters to the rows it is possible to trans-

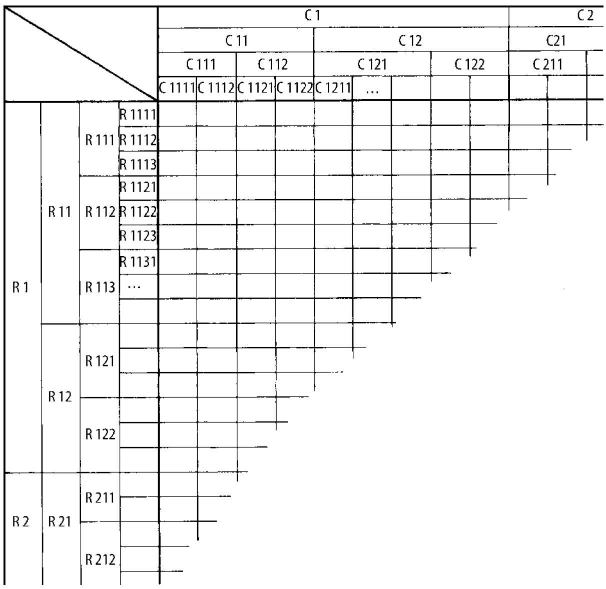


Figure 3.15. Classification scheme with further subdivision of parameters. After [3.10]

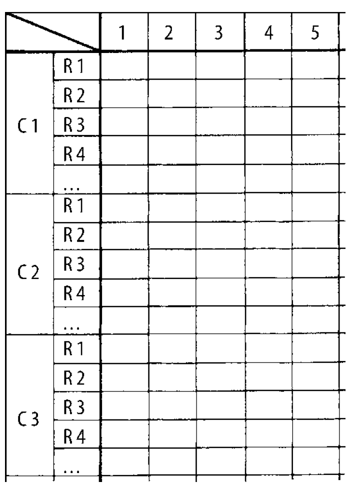


Figure 3.16. Modified classification scheme. After [3.10]

form every classification scheme based on row and column into a scheme in which only the row parameters are retained, and the columns are merely numbered (see Figure 3.16).

Such classification schemes help the design process in a great many ways. In particular, they can serve as design catalogues during all phases of the search for a solution, and they can also help in the combination of subsolutions into overall solutions (see Section 3.2.4). Zwicky [3.77] has referred to them as "morphological matrices".

The choice of classifying criteria or their parameters is of crucial importance. In establishing a classification scheme it is best to use the following step-by-step procedure:

Step 1: Solution proposals are entered in the rows in random order.

Step 2: These proposals are analysed in the light of the main headings (characteristics), such as type of energy, working geometry, working motion, etc.

Step 3: They are classified in accordance with these headings.

The criteria and their parameters can also be obtained from an earlier use of intuitive methods to analyse known solutions or solution ideas.

This procedure not only helps with the identification of compatible combinations but, more importantly, encourages the opening up of the widest possible solution

Classifiing criteria:

Types of energy, physical effects and manifestations

Headings	*Examples*
Mechanical	Gravitation, inertia, centripetal force
Hydraulic	Hydrostatic, hydrodynamic
Pneumatic	Aerostatic, aerodynamic
Electrical	Electrostatic, electrodynamic, inductive, capacitative, piezo-electric, transformation, rectification
Magnetic	Ferromagnetic, electromagnetic
Optical	Reflection, refraction, diffraction, interference, polarisation, infra-red, visible, ultra-violet
Thermal	Expansion, bimetal effect, heat storage, heat transfer, heat conduction, heat insulation
Chemical	Combustion, oxidation, reduction, dissolution, combination, transformation, electrolysis, exothermic and endothermic reactions
Nuclear	Radiation, isotopes, source of energy
Biological	Fermentation, putrefaction, decomposition

Figure 3.17. Classifying criteria and headings (characteristics) for variation in the physical search area

fields. The classifying criteria and characteristics listed in Figures 3.17 and 3.18 can be useful when searching systematically for solutions and the variation of solution ideas for technical systems. They refer to types of energy, physical effects, manifestations, as well as the characteristics of the working geometry, working motions, and the basic material properties (see Section 2.1.4).

Figure 3.19 provides a simple example of searching for a solution to satisfy a subfunction. Here the answer was obtained by varying the type of energy against a number of working principles.

Classifying criteria

Working geometry, working motions and basic material properties

Working geometry

Headings	*Examples*
Type	Point, line, surface, body
Shape	Curve, circle, ellipse, hyperbola, parabola Triangle, square, rectangle, pentagon, hexagon, octagon Cylinder, cone, rhomb, cube, sphere Symmetrical, asymmetrical
Position	Axial, radial, tangential, vertical, horizontal Parallel, sequential
Size	Small, large, narrow, broad, tall, low
Number	Undivided, divided Simple, double, multiple

Working motions

Headings	*Examples*
Type	Stationary, translational, rotational
Nature	Uniform, non-uniform, oscillating Plane or three-dimensional
Direction	In x,y,z direction and/or about x,y,z axis
Magnitude	Velocity
Number	One, several, composite movements

Basic material properties

Headings	*Examples*
State	Solid, liquid, gaseous
Behaviour	Rigid, elastic, plastic, viscous
Form	Solid body, grains, powder, dust

Figure 3.18. Classifying criteria and headings (characteristics) for variation in the form design search area

Type of energy / Working principle	mechanical	hydraulic	electrical	thermal
1	Pot. energy	Liquid reservoir (pot. energy)	Battery	Mass
2	Moving mass (transl.)	Flowing liquid	Capacitor (electr. field)	Heated liquid
3	Flywheel (rot.)			Superheated steam
4	Wheel on inclined plane (rot. + transl. + pot.)			
5	Metal spring	Other springs (compression of fluid + gas) $\Delta p; \Delta V$		
6		Hydraulic reservoir a. Bladder b. Piston c. Membrane (Pressure energy)		

Figure 3.19. Different working principles that satisfy the function "store energy" obtained by varying the type of energy

Figure 3.20 is an example of variation based on working motions.

Figure 3.21 shows the variation in the working geometry in the design of shaft–hub connections. Thanks to such arrangements, the multiplicity of solutions obtained, for instance by the method of forward steps (see Section 2.2.5 and Figure 2.21), can be put into order and completed.

To sum up, the following recommendations are given:

- Classification schemes should be built up step-by-step and as comprehensively as possible. Incompatibilities should be discarded, and only the most promising solution proposals pursued. In so doing, designers should try to determine which classifying criteria contribute to the discovery of a solution, and to examine further variations by modifying the parameters.
- The most promising solutions should be chosen and labelled using a special selection procedure (see Section 3.3.1).
- If possible, the most comprehensive classification schemes should be drawn up (those schemes intended for repeated use), but systems should never be built for the sake of systematics alone.

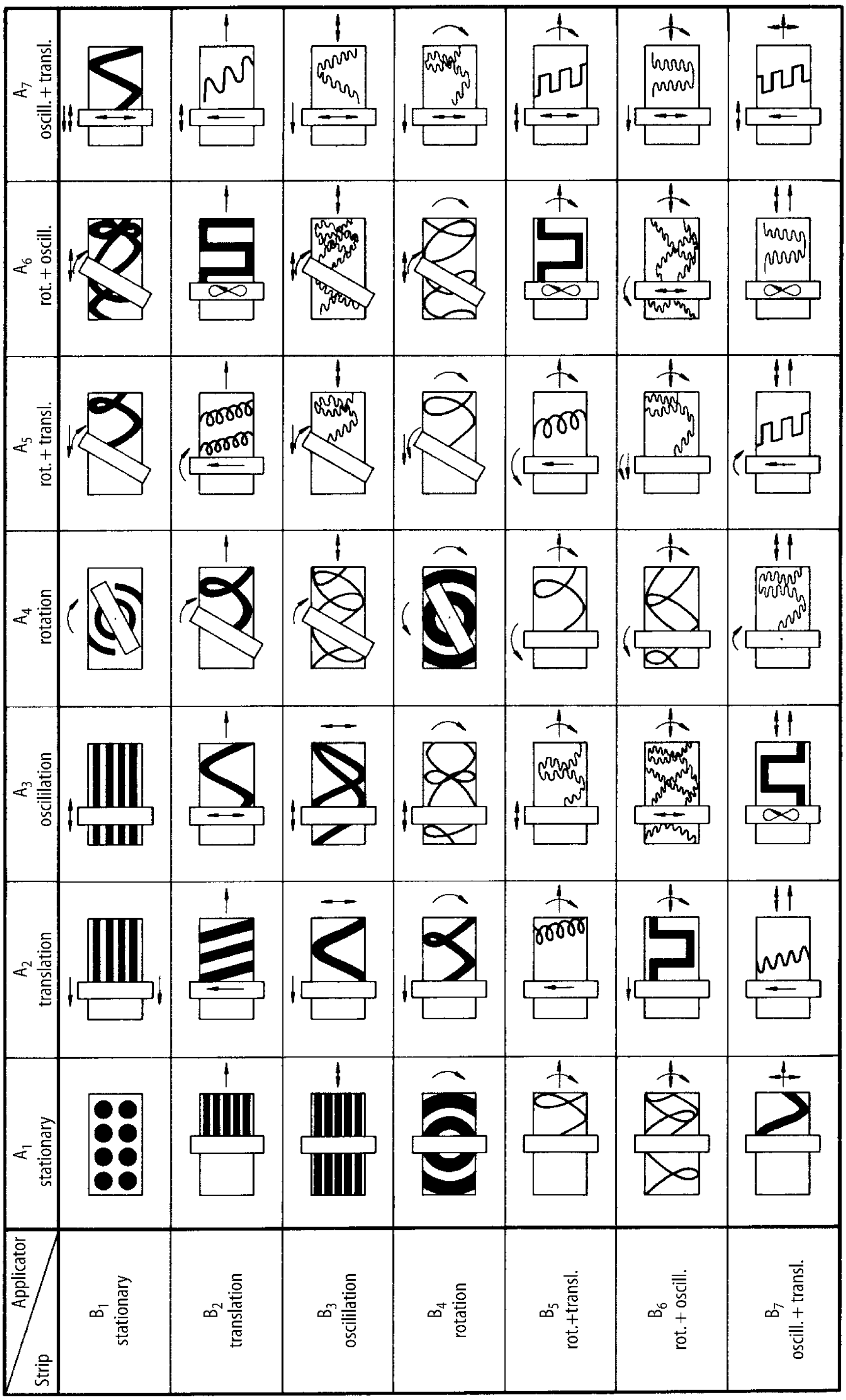


Figure 3.20. Means of coating the backs of carpets by combining the motions of the carpet (strip) and those of the applicator

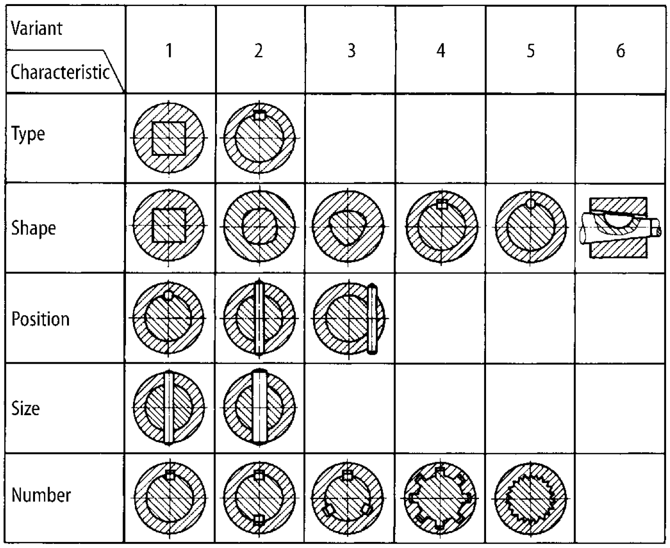


Figure 3.21. Variations in the working geometry for shaft–hub connections

3. Use of Design Catalogues

Design catalogues are collections of known and proven solutions to design problems. They contain data of various types and solutions at distinct levels of embodiment. Thus they may cover physical effects, working principles, principle solutions, machine elements, standard parts, materials, bought-out components, etc. In the past, such data were usually found in textbooks and handbooks, company catalogues, brochures and standards. Some of these contained, apart from purely objective data and suggested solutions, examples of calculations, solution methods and other design procedures. It is also possible to imagine catalogue-like collections for methods and procedures.

Design catalogues should provide:

- quicker, more problem-oriented access to the accumulated solutions or data
- the most comprehensive range of solutions possible, or, at the very least, the most essential ones, which can be extended later
- the greatest possible range of interdisciplinary applications
- data for conventional design procedures as well as for computer-aided methods.

The construction of design catalogues has been studied, above all, by Roth and collaborators [3.62]. Roth suggests that a design catalogue of the type shown in Figure 3.22 is most likely to satisfy all of the demands listed above.

The classifying criteria determine the structure of the catalogue. They influence the ease with which catalogues can be handled and reflect the level of complexity of particular solutions, as well as their degree of embodiment. In the conceptual design phase, for instance, it is advisable to select as classifying criteria the

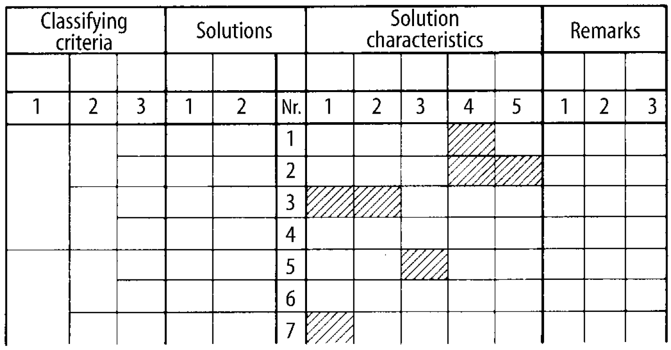


Figure 3.22. Basic structure of a design catalogue. After [3.62]

functions to be fulfilled by the solutions. This is because the conceptual design is based on the underlying subfunctions. When classifying characteristics it is best to choose generally valid functions (see Section 2.1.3), which help to elicit the most product-independent solutions.

Further classifying criteria might include the types and characteristics of energy (mechanical, electrical, optical, etc.), of material or signals, of working geometries, of working motions and of basic material properties. In the case of design catalogues intended for the embodiment design phase, useful classifying criteria include the properties of materials and the characteristics of particular machine elements, such as types of coupling.

The solution column is the main part of the catalogue and contains the solutions. Depending on the level of abstraction, the solutions are represented as sketches, with or without physical equations, or as more or less complete drawings or illustrations. The type and completeness of the information given once again depends on the intended application. It is important that all data is of the same level of abstraction and omits side issues.

The column covering the solution characteristics is important for the choice of solutions.

The remarks column can be used for information about the origin of the data and for additional comments.

The characteristics used for selection may involve a great variety of properties—for instance typical dimensions, reliability, response, number of elements, etc. They help designers in the preliminary selection and evaluation of solutions and, in the case of computer-based catalogues, they can also be used in the final selection and evaluation.

Another important requirement of design catalogues is that they should have uniform and clear definitions and symbols.

The more concrete and detailed the stored information, the more direct but also the more limited the application of a catalogue. With increasing degree of embodiment, data for a given solution become more comprehensive. However, the chances of arriving at a complete solution field decreases because the number of details, for example embodiment variants, increases rapidly. Thus, it may be

Table 3.2. Available design catalogues

Application	Object	Author and reference
General	Construction of catalogues	Roth [3.62]
	List of available catalogues and solutions	Roth [3.62]
Principle solutions	Physical effects	Roth [3.62]
	Solutions to functions	Koller [3.39]
Connections	Types of connections	Roth [3.62]
	Connections	Ewald [3.14]
	Fixed connections	Roth [3.62]
	Wedded joints for steel profiles	Wölse and Kastner [3.75]
	Riveted joints	Roth [3.62], Kopowski [3.41], Grandt [3.26]
	Adhesive joints	Fuhrmann and Hinterwalder [3.18]
	Clamping elements	Ersoy [3.13]
	Principles of threaded joints	Kopowski [3.41]
	Threaded fasteners	Kopowski [3.41]
	Elimination of backlash in threaded joints	Ewald [3.14]
	Elastic joints	Gießner [3.24]
	Shaft–hub connections	Roth [3.62], Diekhöner and Lohkamp [3.9], Kollmann [3.40]
Guides and bearings	Linear guides	Roth [3.62]
	Rotational guides	Roth [3.62]
	Plain and roller bearings	Diekhöner [3.8]
	Bearings and guides	Ewald [3.14]
Power generation, power transmission	Electric motors (small)	Jung and Schneider [3.32]
	Drives (general)	Schneider [3.65]
	Power generators (mechanical)	Ewald [3.14]
	Effects to generate power	Roth [3.62]
	Single-stage power multiplication	Roth [3.62], VDI 2222 [3.70]
	Lifting mechanisms	Raab and Schneider [3.57]
	Screw drives	Kopowski [3.41]
	Friction systems	Roth [3.62]
Kinematics, mechanisms	Solving motion problems using mechanisms	VDI 2727, part 2 [3.72]
	Chain drives and mechanisms	Roth [3.62]
	4-bar mechanisms	VDI 2222, part 2 [3.70]
	Logical inverse mechanisms	Roth [3.62]
	Logical conjunctive and disjunctive mechanisms	Roth [3.62]
	Mechanical flip-flops	Roth [3.62]
	Mechanical non-return safety devices	Roth [3.62], VDI 2222, part 2 [3.70]
	Lifting mechanisms	Raab and Schneider [3.57]
	Uniform-motion transmissions	Roth [3.62]
	Handling devices	VDI 2740 [3.73]
Gearboxes	Spur gears	VDI 2222, part 2 [3.70], Ewald [3.14]
	Mechanical single-stage gearboxes with constant gear ratio	Diekhoner and Lohkamp [3.9]
	Elimination of backlash in spur gears	Ewald [3.14]
Safety technology	Danger situations	Neudorfer [3.52]
	Protective barriers	Neudorfer [3.53]
Ergonomics	Indicators, controls	Neudorfer [3.51]
Production processes	Casting	Ersoy [3.13]
	Drop forging	Roth [3.62]
	Press forging	Roth [3.62]

Function	Input	Output	Physical effects						
E_{mech} → E_{mech}	Force, pressure, torque	Length, angle	Hooke (Tension/ compression/ bending)	Shear, torsion	Upthrust Poisson´s effect	Boyle-Mariotte	Coulomb I and II	...	...
		Speed	Energy Law	Conservation of momentum	Conservation of angular momentum	...	...	...	...
		Accele-ration	Newton´s Law	...	...	...	...	...	...
	Length, angle	Force, pressure, torque	Hooke	Shear, torsion	Gravity	Upthrust	Boyle-Mariotte	Capillary	
			Coulomb I and II	...	...	...	...	...	...
	Speed		Coriolis force	Conservation of momentum	Magnus-effect	Energy law	Centrifugal force	Eddy current	
	Accelera-tion		Newton´s Law	...	...	...	...	...	...
E_{mech} → E_{hyd}	Force, length, speed, pressure	Speed, pressure	Bernoulli	Viscosity (Newton)	Torricelli	Gravitatio-nal pressure	Boyle-Mariotte	Conservation of momentum	...
E_{hyd} → E_{mech}	Speed	Force, length	Profile lift	Turbulence	Magnus-effect	Flow resistance	Back pressure	Reaction principle	...
E_{mech} → E_{therm}	Force, speed	Temperature, quantity of heat	Friction (Coulomb)	1st law	Thomson-Joule	Hysteresis (damping)	Plastic deformation	...	...
E_{therm} → E_{mech}	Temperature, heat	Force, pressure, lenght	Thermal expansion	Steam pressure	Gas Law	Osmotic pressure	...	...	...
E_{elektr} → E_{mech}	Voltage, current, magn. field	Force, speed, pressure	Biot-Savart-effect	Electro-kinetic effect	Coulomb I	Capacitance effect	Johnsen-Rhabeck-effect	Piezoeffect	...
E_{mech} → E_{elektr}	Force, length, speed, pressure	Voltage, current	Induction	Electro-kinetics	Electro-dynamic effect	Piezoeffect	Frictional electricity	Capacitance effect	...
E_{elektr} → E_{therm}	Voltage, current	Temperature, heat	Joule heating	Peltier-effect	Electric arc	Eddy current	...	...	...
E_{therm} → E_{elektr}	Temperature, heat	Voltage, current	Electr. conduction	Thermo-effect	Thermionic emission	Pyroelectri-city	Noise-effect	Semiconductor, Super-conductor	...
E_{mech} → E_{mech}	Force, length, pressure, speed	Force, length, pressure, speed	Lever	Wedge	Poisson´s effect	Friction	Crank	Hydraulic effect	...
E_{hyd} → E_{hyd}	Pressure, speed	Pressure, speed	Continuity	Bernoulli	...	...	...	...	...
E_{therm} → E_{therm}	Temperature, heat	Temperature, heat	Heat conduction	Convection	Radiation	Condensa-tion	Evaporation	Freezing	...
E_{elektr} → E_{elektr}	Voltage, current	Voltage, current	Transformer	Valve	Transistor	Transducer	Thermogal-vanometer	Ohm´s law	...
...	...	...	...	...	...	...	...	...	...

Figure 3.23. Design catalogue of physical effects based on [3.39, 3.48] for the generally applicable functions "change energy" and "vary energy component". Also applicable to flow of signals

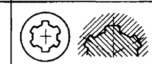
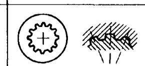
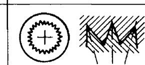
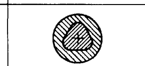
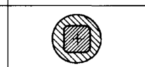
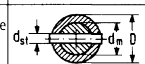
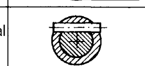
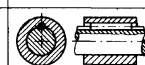
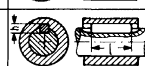
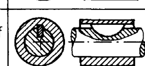
Classifying criteria		Solutions				Solution characteristics															Remarks		
Type of interface	Type of force transmission	Equation	Name	Configuration		Transferable torque	Torque transmission depending on	Axial forces	Stress concentration	Applicable for	Behaviour at overload	Centering possible	Unbalanced force	Axial displacement of hub	Hub moveable	Joint adjustable	Shaft diameter (mm)	Material	Manufacturing effort	Assembly effort	Standard (DIN)	Application examples	Remarks
1	2	1	2	3	Nr.	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17
Normal (form fit)	Direct	$T \leq K \cdot \frac{d}{2} \cdot A_s \cdot \tau_{max}$ or $T \leq K \cdot \frac{d}{2} \cdot A_p \sigma_{max}$	spline shaft		1	large	h, i, l; shaft diameter, (form) factor, centering factor, material	—	large	pulsating or alternating	fracture	yes	no	clearance fit	yes	no	10 – 150	shaft: 37 Cr 4, 41 Cr 4, 42 CrMo	high	small	5461/63 5471/72	toothed wheels	exterior, flank, interior centering possible
			involute spline shaft		2			—													5480, 5482		
			serrated shaft		3			—	medium								150 – 500				5481		short hub possible
			3-sided polygon-shaft		4		e, i, l	—		pulsating or alternating		self-centering	no	clearance fit without load		yes for taper	10 – 100		small, special machinery necessary		—		used for short and thin hubs, conical shaft end possible, broaching or grinding necessary
			4-sided polygon-shaft		5			—									10 – 100				—		
	Indirect		transverse pin		6	small	d_p, D	are supported	large	—		yes	yes	—		yes for taper pin	0.5 – 50	pin: 4D, 5S 6S, 8G 9S, 20K St 50 K St 70 St 60	medium	medium	1, 7 1470 – 77 1481, 6324, 7346	power stretcher, machine tools, vehicles	taper and grooved pin possible
			tangential pin		7					—				—	no								
			in line pin		8		d_p, l	—	medium	—										small			
			key joint		9		h, l, b	—	large	—				clearance fit		no	5 – 500	spring St 60; shaft, hub: GG, GS ST			6885		
			Woodruff key		10			—		—											6888		

Figure 3.24. Extract of a catalogue for shaft–hub connections. After [3.62]

possible to provide a full list of physical effects fulfilling the function "channel", but it would hardly be possible to list all of the potential embodiments of bearings (channelling a force from a rotating to a stationary system).

Table 3.2 lists the currently available design catalogues that satisfy the requirements and structure described above. Therefore, in what follows we include just a few examples of, or extracts from, available design catalogues.

Figure 3.23 shows a catalogue of physical effects associated with the functions "change energy" and "vary energy component". It is based on Koller [3.39] and Krumhauer [3.48]. The catalogue makes it possible to derive these effects from the classifying criteria, that is, the "inputs and outputs" of the functions. The characteristics on which the selection is based must be derived from the technical literature.

Figure 3.24 shows an extract of a catalogue for shaft–hub connections based on [3.62]. In this, unlike the previous catalogue, the solutions are concrete enough, thanks to the specification of the form design features, for the embodiment design phase to start with a scale layout drawing.

Computer-based systems are used to facilitate searching through catalogues, company brochures, supplier information and other documents. Hypermedia software provides a way of structuring, storing and retrieving the contents of such documents. It allows the flexible manipulation of chunks of information, and the representation and linking of objects and procedures in a specific knowledge domain, using different representation principles. This is called navigating in a hypermedia system [3.58]. To use distributed sources of information, a global network is required, such as the internet (www). Using the internet, so-called "virtual markets" or "virtual supply chains" can be created with which designers can communicate from their work places [3.4].

3.2.4 Methods for Combining Solutions

As described in Sections 2.1.3 and 2.2.5, it is often useful to divide problems, tasks and functions into subproblems, subtasks and subfunctions and to solve these individually (factorisation method) (see also Section 6.3). Once the solutions for subproblems, subtasks or subfunctions are available, they have to be combined in order to arrive at an overall solution.

The methods we have been describing, particularly those with an intuitive bias, may have led to the discovery of suitable combinations. However, there are special methods for arriving at such syntheses more directly. In principle, they must permit a clear combination of solution principles with the help of the associated physical and other quantities and the appropriate geometrical and material characteristics. When analysing combinations that involve software elements, it is important to identify and use appropriate solution characteristics.

The main problem with such combinations is ensuring the physical and geometrical compatibility of the solution principles to be combined, which in turn ensures the smooth flow of energy, material and signals, and avoids geometrical interference in mechanical systems. For information systems, the main problem is the compatibility requirements of the information flow.

A further problem is the selection of technically and economically favourable combinations of principles from the large field of theoretically possible combinations. This aspect will be discussed at greater length in Section 3.3.1.

1. Systematic Combination

For the purpose of systematic combination, the classification scheme to which Zwicky [3.77] refers as the "morphological matrix" (see Figure 3.25) is particularly useful. Here, the subfunctions, usually limited to the main functions, and appropriate solutions (solution principles) are entered in the rows of the scheme.

If this scheme is to be used for the elaboration of overall solutions, then at least one solution principle must be chosen for every subfunction (that is, for every row). To provide the overall solution, these principles (subsolutions) must then be combined systematically into an overall solution. If there are m_1 solution principles for the subfunction F_1, m_2 for the subfunction F_2, and so on, then after complete combination we have $N = m_1 \cdot m_2 \cdot m_3 \cdots m_n$ theoretically possible overall solution variants.

The main problem with this method of combination is to decide which solution principles are compatible; that is, to narrow down the theoretically possible search field to the practically possible search field.

The identification of compatible subsolutions is facilitated if:

- the subfunctions are listed in the order in which they occur in the function structure, if necessary separated according to flow of energy, material and signals
- the solution principles are suitably arranged with the help of additional column parameters, for example the types of energy
- the solution principles are not merely expressed in words but also in rough sketches

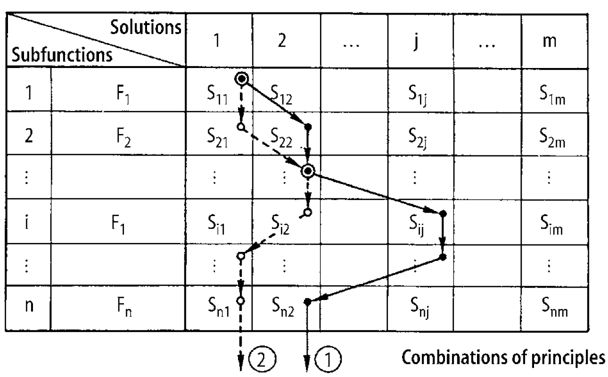


Figure 3.25. Combining solution principles into combinations of principles: Combination 1: $S_{11} + S_{22} + \dots S_{n2}$; Combination 2: $S_{11} + S_{21} \dots S_{n1}$

- the most important characteristics and properties of the solution principles are recorded as well.

The verification of compatibilities, too, is facilitated by classification schemes. If two subfunctions to be combined—for instance, "change energy" and "vary mechanical energy component"—are entered respectively in the column and row headings of a matrix with their characteristics in the appropriate cells, then the compatibility of the subsolutions can be verified more easily than it could be if such examinations were to be confined to the designer's head. Figure 3.26 illustrates this type of compatibility matrix. Further examples of this method of combination will be found in Section 6.4.2 (Figures 6.15 and 6.19).

To sum up:

- Combine only compatible subfunctions.
- Pursue only such solutions as meet the demands of the requirements list and fall within the available resources (see selection procedures in Section 3.3.1).
- Concentrate on promising combinations and establish why these should be preferred above the rest.

In conclusion, it must be emphasised that what we have been discussing is a generally valid method of combining subsolutions into overall solutions. The method can be used for the combination of working principles during the conceptual phase, and of subsolutions or even of components and assemblies during the embodiment phase. Because it is essentially a method of information processing, it

Change energy / Vary mechan. energy component		Electric motor	Oscillating solenoid	Bimetal spiral in hot water	Oscillating hydraulic piston	...
		1	2	3	4	...
Four-bar linkage	A	if A capable of rotating	slow motion	yes	additional lever linkage but only for low piston speeds	...
Chain drive Spur gear drive	B	yes	slow rotation only through additional elements (free wheeling etc.), difficult to reverse direction	gear segments suffice, depending on angle of rotation	with a rack and swivel, but only for low piston speeds	...
Maltese drive	C	yes look out for shock loads	see B2	yes (when angle of rotation is small lever with sliding block)	lever with sliding block, but only for low piston speeds	...
Friction wheel drive	D	yes	see B2	large forces because of torque during slow movement, imprecise positioning	see D3	...
...	...	...	...	...	...	...

☒ very difficult to apply (do not pursue further)

☒ can only be applied under certain circumstances (defer)

Figure 3.26. Compatibility matrix for combination possibilities of the subfunctions "change energy" and "vary mechanical energy component". After [3.10]

is not confined to technical problems but can also be used in the development of management systems and in other fields.

2. Combining With the Help of Mathematical Methods

Mathematical methods and computers should only be used for the combination of solution principles if real advantages can be expected from them. Thus, at the relatively abstract conceptual phase, when the nature of the solution is not yet fully understood, a quantitative elaboration—that is, a mathematical combination along with an optimisation—is quite out of place and can be misleading. The exceptions are combinations of known elements and assemblies, for instance in variant or circuit design. In the case of purely logical functions, combinations can be performed with the help of Boolean algebra [3.17, 3.59] in, say, the layout of safety systems or the optimisation of electronic or hydraulic circuits.

In principle, the combination of subsolutions into overall solutions with the help of mathematical methods calls for knowledge of the characteristics or properties of the subsolutions that are expected to correspond with the relevant properties of the neighbouring subsolutions. These properties must be unambiguous and quantifiable. In the formation of principle solutions (for example working structures), data about the physical relationships may be insufficient, since the geometrical relationships may have a limiting effect and hence may, in certain circumstances, lead to incompatibilities. In that case, physical equations and geometrical structure must first be matched mathematically, and this is not generally possible except for systems of low complexity. For systems of higher complexity, in contrast, such correlations often become ambiguous, so that designers must once again choose between variants. We may, accordingly, speak of dialogue systems in which the process of combination consists of mathematical and creative steps.

This makes it clear that, with increasing physical realisation or embodiment of a solution, it becomes simpler to establish quantitative combination rules. However, the number of properties increases and with them the number of constraints and optimisation criteria, so that the mathematical effort becomes very great and requires computer support.

3.3 Selection and Evaluation Methods

3.3.1 Selecting Solution Variants

For the systematic approach, the solution field should be as wide as possible. By considering all possible classifying criteria and characteristics, designers are often led to a larger number of possible solutions. This profusion constitutes the strength and also the weakness of the systematic approach. The very great theoretically admissible, but practically unattainable, number of solutions must be reduced at the earliest possible moment. On the other hand, care must be taken not to eliminate valuable working principles, because it is often only through their combination